

॥ श्री गणेशाय नमः ॥

Co-Ride – A Carpooling Solution

An Engineering Project in Community Service

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in partial fulfillment of the requirements for the degree of

Bachelor of Engineering and Technology



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ABSTRACT

In this report we have presented a design and how to implement the car-pooling system or car ride sharing system among the common people. Our system works in real time and it is dynamic in nature.

Our System matches the rides of many users in one way and it helps in occupying the vacant seats in the longest common ride. Our system is interactive with the user unlike the previous system which were static and not interactive and easy to use. Old systems don't contain information about the exact location of pickup of a customer and also the dropping point location of a customer.

Our system is based on real time updates in which the exact location of pick up and drop off point is known to the driver with the help of advancement in technology like exact location based on GPS reading and GIS system coordinates. The interaction among the different type of users and the route matching algorithm enables the user as well as system to match the suitable ride share according to different users in the way.

Our system builds the new routes from more than two routes dynamically with attention to the customer and it prefers maximum deviation from the original path of ride in order to catch more customers in the single ride on the way.

We believe that the growing attention towards the increase in environmental conditions such as pollution level and other issues like increase in the density and volume of transportation system in the cities as well as villages. These types of system are going to save the people time, money and other things that are beneficial to the human beings that are living in the system surroundings.

INDIVIDUAL CONTRIBUTION

(Harshil Patel)

User Interface (Code(XML)) -

In mobile application development, UI (User Interface) coding involves implementing the visual and interactive elements of the app's user interface. The components of UI coding in mobile application development include:

Layouts and Views:

- Layout Managers: Classes or components that determine the arrangement of UI elements on the screen, such as LinearLayout, RelativeLayout, ConstraintLayout, etc.
- Views: User interface elements like buttons, text fields, labels, images, checkboxes, radio buttons, etc., which are defined in XML layout files or created programmatically.

Styling and Theming:

- Styles and Themes: Defining styles and themes to apply consistent visual attributes (such as colours, fonts, padding, margins, etc.) across UI elements and screens.
- Resources: Utilizing resource files (e.g., XML files in Android) to manage styles, themes, colours, dimensions, and other UI-related attributes.

User Input Handling:

- Event Handlers: Implementing event listeners and handlers to capture user interactions such as clicks, taps, swipes, and gestures on UI elements.
- Input Validation: Validating user input entered through text fields, checkboxes, radio buttons, etc., to ensure data integrity and prevent errors.

Animations and Transitions:

- Property Animations: Animating UI elements to create visual effects like fading, scaling, rotating, translating, etc., using property animation APIs.
- Transition Animations: Defining transition animations for smooth screen transitions, such as slide, fade, explode, and shared element transitions.

Adaptive UI Design:

- Responsive Layouts: Developing UI layouts that adapt to different screen sizes, resolutions, orientations, and aspect ratios to ensure consistent user experience across devices.

Information Architecture:

- Content Organization: Structuring and organizing app content, features, and functionalities in a logical hierarchy to facilitate navigation and discovery.

Layout and Navigation:

- Screen Layout: Arrangement of elements on the screen, including headers, footers, menus, and content areas.
- Navigation Patterns: Design of navigation menus, tabs, buttons, and gestures for seamless movement between screens and sections within the app.
- Back Button: Placement and functionality of the back button for easy navigation within the app.

Visual Design:

- Colour Scheme: Selection of colours for backgrounds, text, buttons, and other UI elements to create a visually appealing and cohesive design.
- Typography: Choice of fonts, sizes, and styles for text elements to enhance readability and convey the app's personality.
- Iconography: Use of icons and visual elements to represent actions, features, and categories within the app.
- Themes: Consistent application of visual themes and styles to maintain brand identity and create a unified user experience.

Interaction Design:

- User Input: Design of input fields, buttons, checkboxes, and other interactive elements for user engagement and data entry.
- Animation: Integration of animations, transitions, and micro-interactions to enhance usability, provide context, and guide user interactions.

Content Presentation:

- Content Hierarchy: Structuring and prioritizing content elements based on importance and relevance to guide user attention and flow.
- Whitespace: Use of whitespace and visual breathing room to improve readability, organization, and clarity of content.

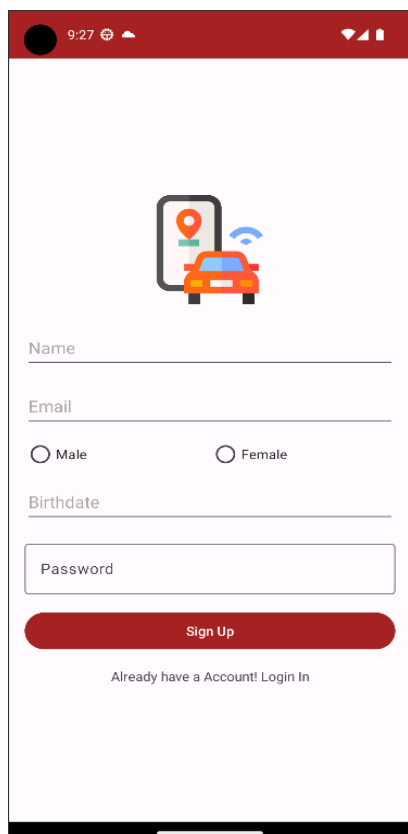
Wireframing:

- Wireframes: Creating low-fidelity visual representations of app screens and layouts, focusing on content placement, information hierarchy, and functionality without detailed visual design.

Visual Design:

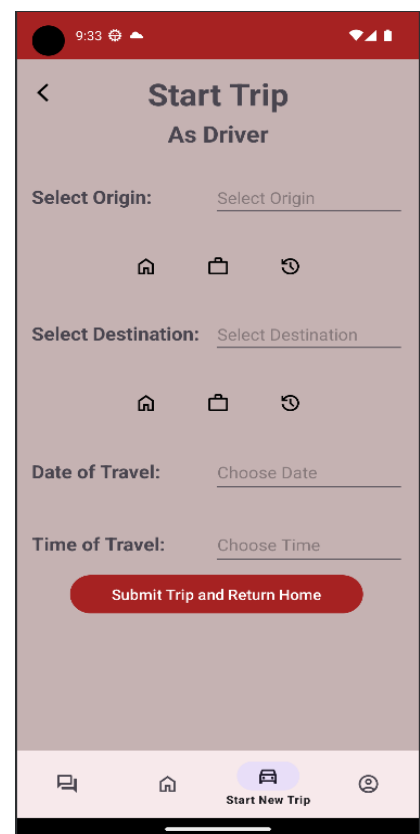
- UI Elements: Designing visually appealing user interface elements, including colours, typography, icons, buttons, and imagery, to create a consistent and engaging visual identity.

- Brand Identity: Incorporating the app's branding elements, such as logos, colour schemes, and brand guidelines, to reinforce brand recognition and identity.



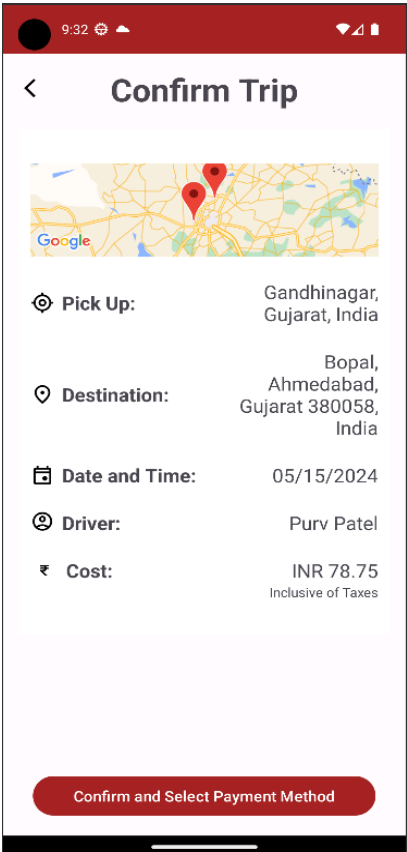
A wireframe of a sign-up page. At the top, there is a status bar with the time 9:27 and battery level. Below the status bar is a red header bar. The main content area is white. It features a central icon of a car with a location pin and a Wi-Fi signal. Below the icon are input fields for Name, Email, Birthdate, and Password. There are radio buttons for Male and Female. A red button labeled 'Sign Up' is at the bottom. Below the button is a link that says 'Already have a Account! Login In'.

Sign Up Page Layout(UI)

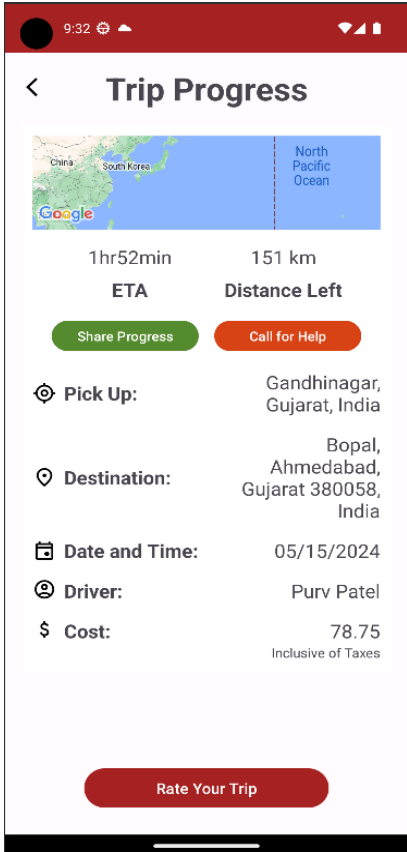


A wireframe of a start trip page. At the top, there is a status bar with the time 9:33 and battery level. Below the status bar is a red header bar. The main content area is white. It features a back arrow and the title 'Start Trip As Driver'. Below the title are input fields for 'Select Origin:' and 'Select Destination:'. There are three icons (home, car, clock) below each input field. Below these are input fields for 'Date of Travel:' and 'Time of Travel:'. A red button labeled 'Submit Trip and Return Home' is at the bottom. Below the button is a bottom navigation bar with four icons: a chat bubble, a home icon, a car icon (highlighted), and a profile icon. The text 'Start New Trip' is below the car icon.

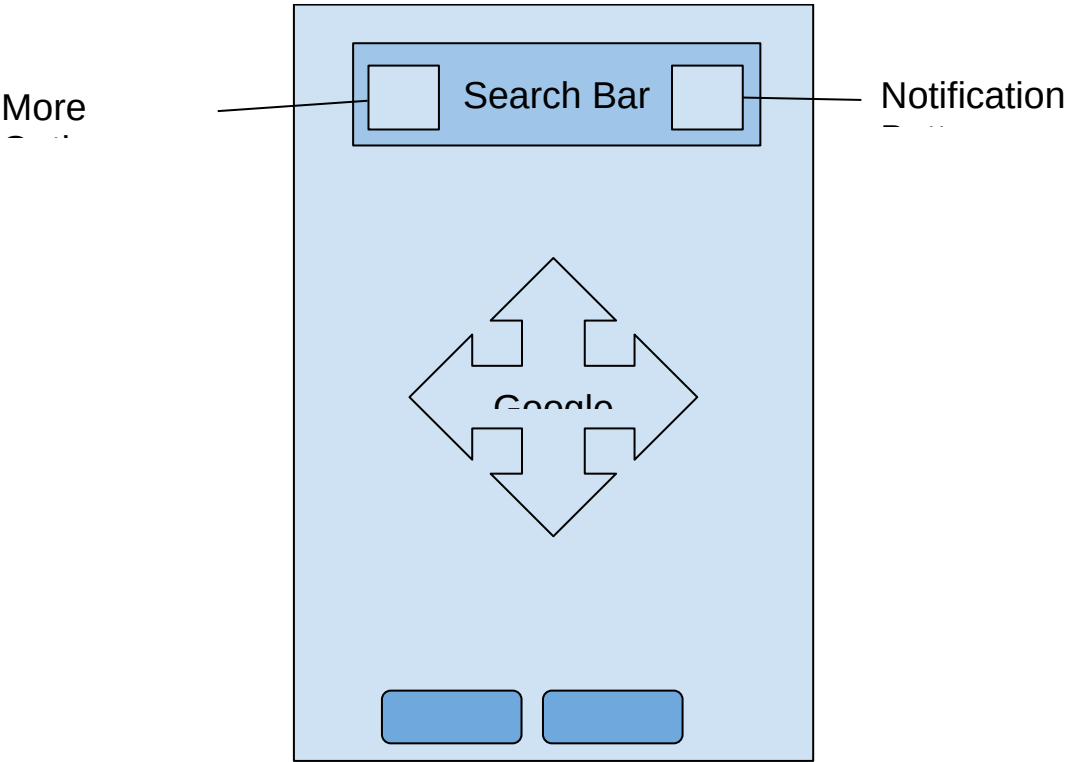
Start Trip Page Layout



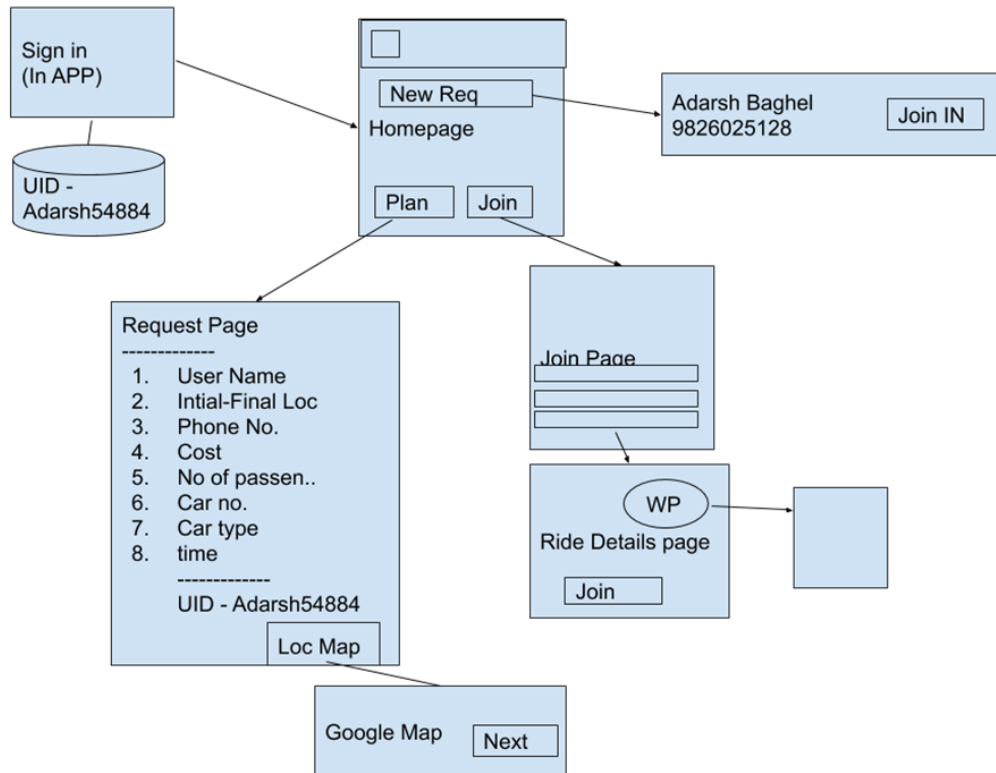
Confirm Trip Page Layout(UI)



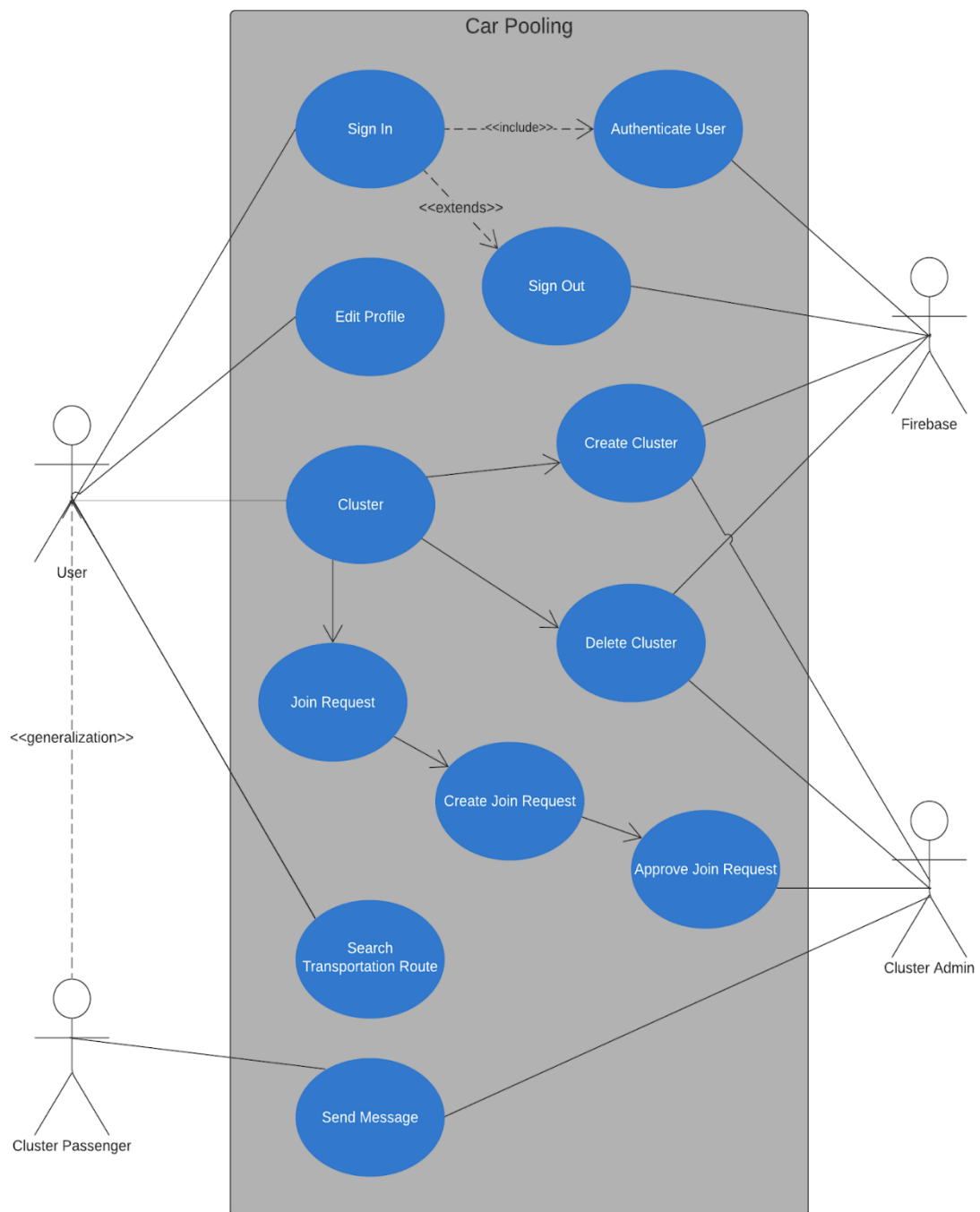
Trip Progress Page(UI)



Home Page Layout



User Interface Design Layout



Flow of User Interface



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UI Coding

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Technical Skills Native Android (Java, Kotlin), Hybrid Android (Flutter, Dart), Firebase, Jetpack Compose, XML, Figma, Ui/UX

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